
Project Documentation

for

Hospital Management System Analysis

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1. Introduction

This project provides a comprehensive analytical study of a hospital management system using seven interconnected datasets: Doctors, Patients, Claims & Billing, Denials, Procedures, Encounters, and Medications. The objective of the analysis is to understand operational performance, medical service utilization, financial outcomes, and patterns across clinical and administrative workflows.

*Data cleaning, transformation, and modeling were conducted to ensure consistency, accuracy, and reliability. Multiple analytical tools—including **Excel, SQL, Python, Power BI, and Tableau**—were utilized to explore the data and generate actionable insights.*

The final deliverables highlight trends in patient encounters, procedure utilization, physician performance, medication patterns, claim outcomes, and denial reasons. The findings support data-driven recommendations aimed at improving operational efficiency, clinical effectiveness, and financial performance within the healthcare environment.

2. Project Objectives

The main objectives of this project are:

1. **Data Quality Improvement:** Perform thorough data cleaning and preprocessing to ensure consistency, accuracy, and completeness across all datasets.
2. **Data Modeling:** Develop relationships between the seven key datasets (Doctors, Patients, Encounters, Procedures, Medications, Claims & Billing, Denials) to support integrated analysis.
3. **Exploratory Data Analysis (EDA):** Identify patterns, trends, and anomalies in hospital operations, clinical services, and financial performance.
4. **KPI Evaluation:** Calculate key performance indicators related to patient care, physician performance, procedures, medications, claims, and denials.
5. **Insights & Recommendations:** Provide actionable insights to improve operational efficiency, clinical effectiveness, and financial outcomes.
6. **Visualization:** Use Power BI, Tableau, and Python visualizations to communicate findings effectively.

3. Dataset Overview

The project utilizes seven interconnected datasets:

1. **Doctors:** Contains information about physician identifiers, specialties, department, and performance metrics.
2. **Patients:** Records demographic details, admission information, and patient identifiers.
3. **Encounters:** Details of patient visits, including visit type, dates, and linked physician.
4. **Procedures:** Information on medical procedures performed during encounters, including procedure codes and costs.
5. **Medications:** Lists prescribed medications, dosage, administration details, and related patient encounters.
6. **Claims & Billing:** Financial records including billed amounts, payments, and submitted claims.
7. **Denials:** Information on claim denials, denial reasons, appeal status, and final outcomes.

This structured dataset allows for comprehensive analysis of hospital operations, clinical services, and financial workflows.

Dataset Source:

The datasets used in this project are publicly available on Kaggle:

<https://www.kaggle.com/datasets/rajkumarpadmanabhan/ca-hospital-dataset-q1-2025>

4. Tools & Technologies Used

The hospital management project was implemented **comprehensively across all seven datasets** using multiple analytical tools, with each tool applied fully to all aspects of the project:

1. **Microsoft Excel:**

Performed full-scale data exploration, cleaning, filtering, pivot tables, and initial visualizations for all datasets (Doctors, Patients, Encounters, Procedures, Medications, Claims & Billing, Denials).

2. **SQL:**

Fully utilized for querying, joining, and transforming relational data across all datasets, supporting complex analysis and data modeling.

3. **Python:**

Applied extensively using libraries such as pandas, numpy, and matplotlib and for advanced data cleaning, exploratory analysis, and visualization across the entire project.

4. **Power BI:**

Interactive dashboards were developed for **all datasets**, allowing stakeholders to track KPIs, trends, and operational metrics across hospital management.

5. **Tableau:**

Advanced visualizations and trend analysis were performed comprehensively on all datasets to uncover patterns, insights, and relationships across clinical and administrative workflows.

Note: Each tool was applied independently and fully to ensure data integrity, comprehensive analysis, and high-quality insights across the hospital management system.

5. Data Cleaning & Preparation

The data cleaning and preprocessing for the hospital management project were performed comprehensively across all datasets (Doctors, Patients, Encounters, Procedures, Medications, Claims & Billing, Denials) using **Power Query in Excel**, ensuring consistency and reliability for further analysis and modeling. The cleaning steps included the following:

1. Column Management:

- Removed unnecessary or irrelevant columns from all datasets to focus on essential attributes for analysis.

2. Missing Values Handling:

- Filled missing values (NULLs) with appropriate replacements, such as “No Claim” for missing claim IDs or the mode for missing city and state entries.

3. Date Formatting:

- Standardized date formats across all datasets to ensure consistent temporal analysis.
- Extracted additional features such as month and year from date fields where relevant.

4. Duplicate Removal:

- Identified and removed duplicate records in all datasets to maintain data integrity.

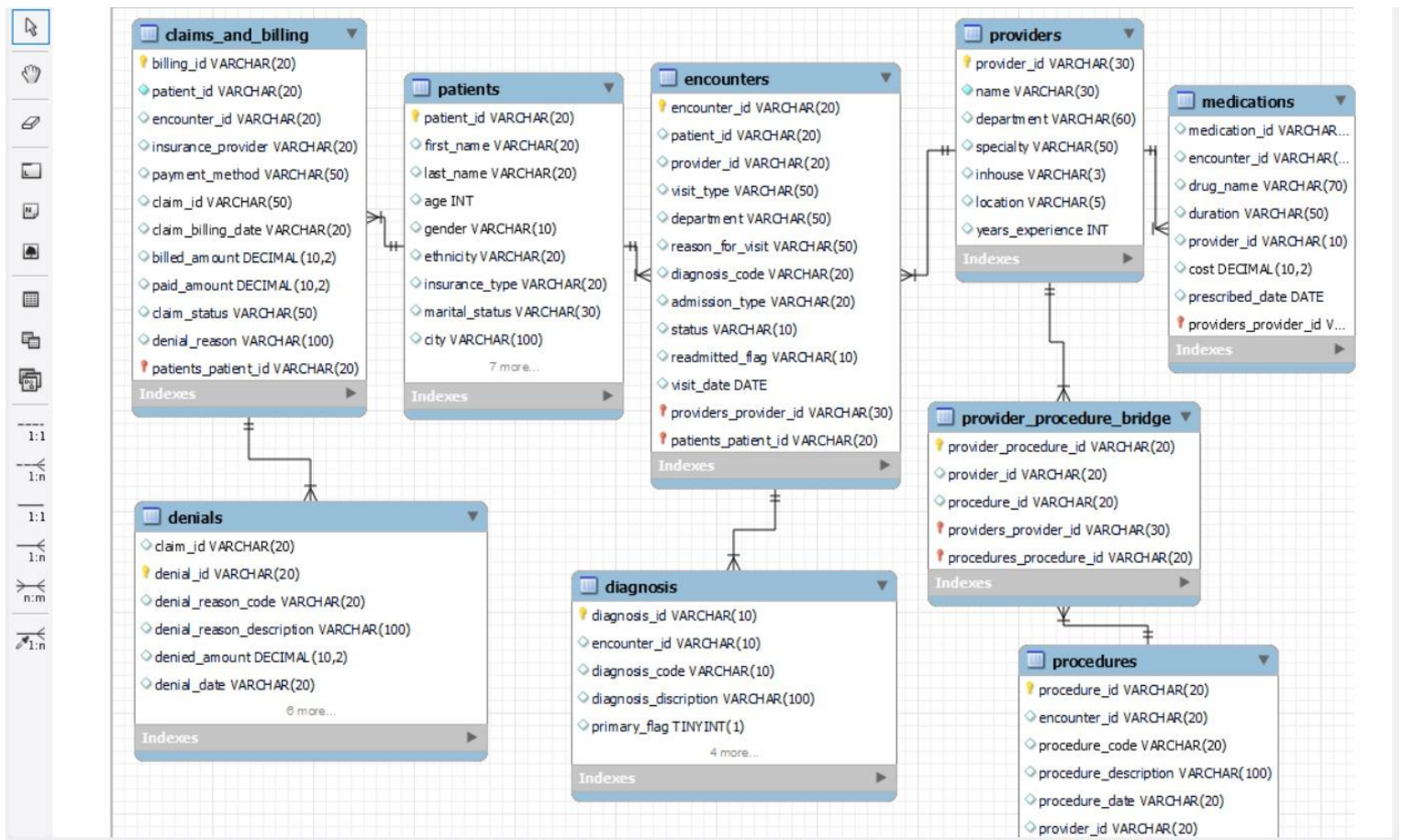
5. Conditional Columns & Derived Features:

- Calculated patient age from date of birth.
- Created age groups using conditional logic.
- Adjusted marital status for patients under 16 years old.

6. Data Modeling & Relationships

6.1 Entity Relationship Diagram (ERD)

The ERD below presents the core entities of the hospital management system and the relationships between patients, encounters, providers, procedures, medications, claims, and denials.



6.2 Model Summary

- *Patients have multiple encounters.*
- *Each encounter may include diagnoses, procedures, and medications.*
- *Providers are linked to encounters and procedures (via a bridge table).*
- *Each encounter generates one claim.*
- *A claim may have multiple denials.*

6.3 Design Goal

The data model is structured to support end-to-end healthcare analytics, including clinical activities, financial outcomes, and claim lifecycle tracking.

7. Exploratory Data Analysis (EDA) & Visualizations

The Exploratory Data Analysis was performed to understand the structure, quality, and behavior of the datasets before modeling and visualization.

The following steps were applied across all tables:

7.1 Data Structure Review

- Inspected the number of rows, columns, and data types.*
- Verified primary identifiers (IDs) and ensured consistency across tables.*

7.2 Missing Values Analysis

- Detected and quantified missing values.*
- Applied appropriate handling based on column type (e.g., replacing missing text values with “No Claim” or using the mode for City/State fields).*

7.3 Duplicate Detection

- Identified and removed duplicate records across multiple tables to ensure data integrity.*

7.4 Data Type Validation

- Ensured correct formatting for dates, numeric values, and categorical fields.*
- Converted all date fields to a consistent format.*

7.5 Statistical Summary

- Reviewed basic statistics (min, max, mean, median) for numerical fields.*
- Assessed distribution of key variables such as billed amount, age, and encounter types.*

7.6 Outlier & Anomaly Checks

- *Detected unusual values (e.g., negative amounts or unrealistic ages).*
- *Verified financial fields such as billed and paid amounts.*

7.7 Key Insights Extraction

- *Identified initial trends, patterns, and potential issues to be investigated further during the modeling and visualization stages.*

8. Performance Metrics & KPIs

The hospital management project measured several key performance indicators (KPIs) to evaluate operational, clinical, and financial performance across the seven datasets. These metrics provide insights for data-driven decision-making and process improvement.

1. Doctors & Encounters:

- *Total number of all and active providers.*
- *Total recorded patient visits*
- *Mean years of professional experience.*
- *Number of unique medical departments*

2. Patients:

- *Overall count of registered patients.*
- *Mean age across all patients*

3. Finance:

- *Total value of all submitted claims*
- *Total paid amount*
- *Total denied amount*

Visualization of KPIs:

- ***Power BI & Tableau Dashboards*** were used to track all KPIs interactively.
- *Python visualizations supported trend analysis, correlations, and outlier detection.*
- *KPIs were applied consistently across all datasets, providing a comprehensive overview of hospital operations, financial efficiency, and clinical performance.*

9. Key Insights & Findings

Doctors & Encounters:

- *Variation in patient load and procedures across doctors and departments.*
- *Uneven workload distribution indicates potential resource bottlenecks.*

Patients:

- *Key age groups identified for targeted healthcare programs.*
- *Younger patients (<16) mostly single; some segments show high repeat encounters.*

Procedures:

- *Most procedures concentrated in few departments.*
- *Significant cost variation suggests inefficiencies.*

Medications:

- *Prescription patterns differ by age and department.*
- *Frequent medications highlight critical treatment areas.*

Claims & Billing:

- *Billing peaks align with high patient volumes.*
- *Cleaning ensured accurate claim analysis; trends reveal revenue patterns.*

Denials:

- *Main reasons: documentation errors and incorrect payer info.*
- *Appeal success varies; trends indicate process inefficiencies.*

10. Recommendations

1. Data Management & Quality:

- *Implement automated validation and standardize data entry to reduce errors and missing values.*

2. Operational Efficiency:

- *Optimize physician schedules and review high-volume departments for adequate staffing and resources.*

3. Clinical Process Improvement:

- *Monitor frequent procedures and medications to reduce unnecessary variations.*
- *Use patient demographics to design targeted healthcare programs.*

4. Financial & Billing Optimization:

- *Train staff on proper documentation and payer requirements.*
- *Automate alerts for incomplete claims and track denial trends for corrective actions.*

5. Visualization & Reporting:

- *Maintain interactive dashboards and generate periodic reports to support decision-making.*

6. Continuous Improvement:

- *Regularly review datasets, KPIs, and dashboards.*
- *Use predictive analytics to anticipate patient load, claims, and resource needs.*

Note: *Implementing these steps will enhance operational efficiency, improve patient care, and optimize financial performance.*

11. Conclusion

The Hospital Management Data Analysis Project provided a comprehensive examination of clinical, operational, and financial aspects of hospital operations across seven interconnected datasets.

Through systematic data cleaning, modeling, and exploratory analysis, critical insights were uncovered regarding physician performance, patient demographics, procedures, medications, billing, and claim denials. Key performance indicators and interactive dashboards facilitated a clear understanding of trends, bottlenecks, and opportunities for improvement.

The findings support actionable recommendations aimed at:

- Enhancing operational efficiency and resource allocation*
- Improving clinical effectiveness and patient care*
- Reducing financial losses and optimizing billing and claims processes*

This project demonstrates the value of a data-driven approach in hospital management, providing a foundation for continuous monitoring, predictive analytics, and informed decision-making in the healthcare environment.